

Ringdown Analysis Using Orthonormal Modes

Based on arXiv:2507.12376 and arXiv:2605.03576

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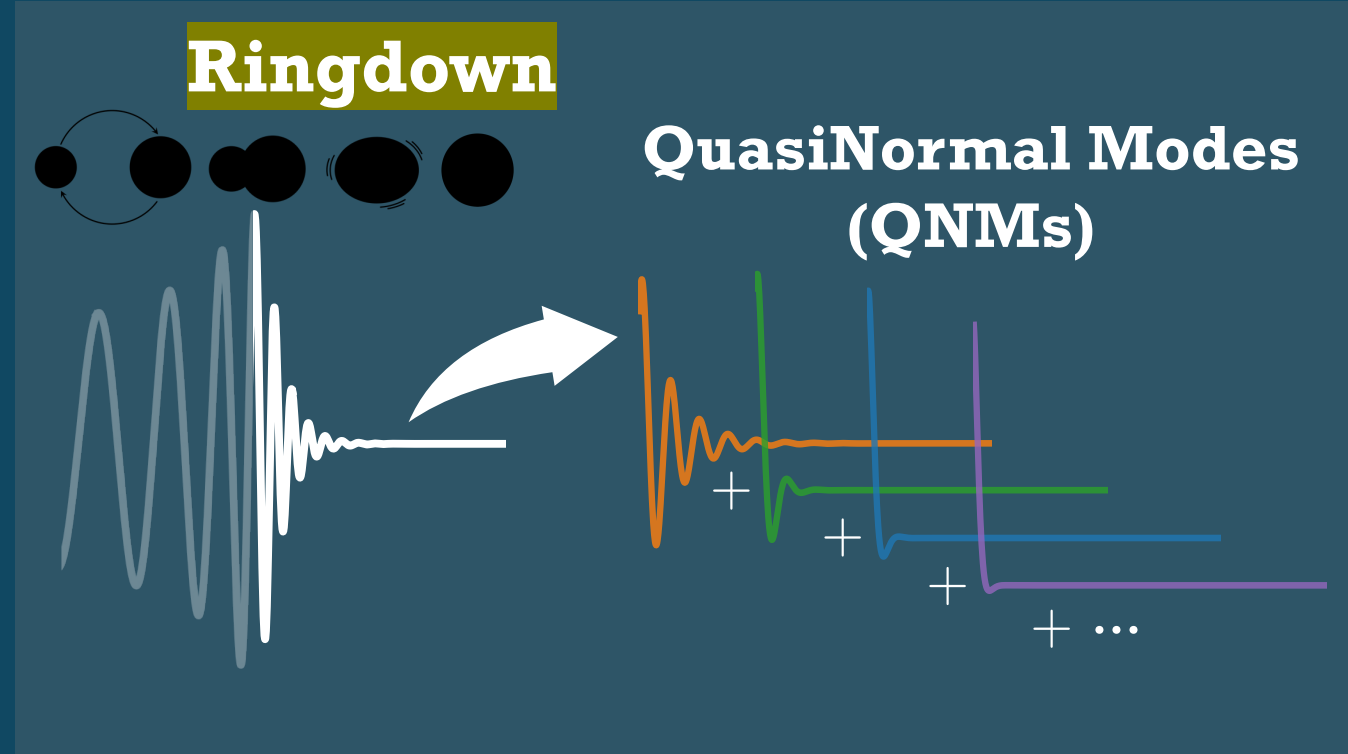
Theoretical Developments in Gravitational Wave Physics
YITP, June 15-19, 2026

Image credit: LIGO/T. Pyle

Ringdown signal can be modeled as a superposition of damped sinusoids:

$$h(t) = h_+(t) - ih_\times(t)$$
$$= \sum_{\ell m n} \left[\mathcal{A}_{\ell m n} e^{-t/\tau_{\ell m n}} e^{-i2\pi f_{\ell m n} t} + \mathcal{A}'_{\ell m n} e^{-t/\tau_{\ell m n}} e^{i2\pi f_{\ell m n} t} \right]$$

- ℓ, m : Angular indices
- n : Overtone number



- $\mathcal{A}_{\ell m n}, \mathcal{A}'_{\ell m n}$: Complex amplitudes, depend on the source properties (free params)
- $f_{\ell m n}, \tau_{\ell m n}$: QNM frequency and damping time, determined by spacetime (mass and spin of the remnant black hole)

- The ringdown signal

$$h(t) = h_+(t) - ih_\times(t) = \sum_{\ell mn} \left[\mathcal{A}_{\ell mn} e^{-t/\tau_{\ell mn}} e^{-i2\pi f_{\ell mn} t} + \mathcal{A}'_{\ell mn} e^{-t/\tau_{\ell mn}} e^{i2\pi f_{\ell mn} t} \right]$$

- The strain measured by the I th detector

$$h^I(t) = F_+^I h_+(t) + F_\times^I h_\times(t) = \sum_{\ell mn} \sum_{j=0}^3 c_{j,\ell mn} v_{j,\ell mn}^I(t)$$

- Basis function
(determined by the mass and spin)

$$v_{0,\ell mn}^I(t) = F_+^I e^{-(t-t_S^I)/\tau_{\ell mn}} \cos[2\pi f_{\ell mn}(t - t_S^I)],$$

$$v_{1,\ell mn}^I(t) = F_+^I e^{-(t-t_S^I)/\tau_{\ell mn}} \sin[2\pi f_{\ell mn}(t - t_S^I)],$$

$$v_{2,\ell mn}^I(t) = F_\times^I e^{-(t-t_S^I)/\tau_{\ell mn}} \cos[2\pi f_{\ell mn}(t - t_S^I)],$$

$$v_{3,\ell mn}^I(t) = F_\times^I e^{-(t-t_S^I)/\tau_{\ell mn}} \sin[2\pi f_{\ell mn}(t - t_S^I)].$$

- Coefficients
(determined by the mode amplitudes)

$$c_{0,\ell mn} = \text{Re}(\mathcal{A}_{\ell mn} + \mathcal{A}'_{\ell mn}),$$

$$c_{1,\ell mn} = \text{Im}(\mathcal{A}_{\ell mn} - \mathcal{A}'_{\ell mn}),$$

$$c_{2,\ell mn} = -\text{Im}(\mathcal{A}_{\ell mn} + \mathcal{A}'_{\ell mn}),$$

$$c_{3,\ell mn} = \text{Re}(\mathcal{A}_{\ell mn} - \mathcal{A}'_{\ell mn}).$$

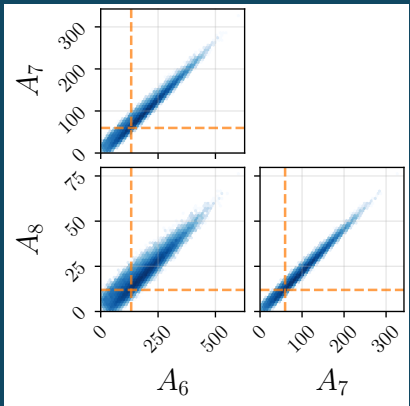
Challenge in Multi-Mode Analysis

4/16

- Detection of multiple QNMs in a single GW event enables tests of GR using only the ringdown signal (BH spectroscopy).
- More QNMs $\longrightarrow$ More stringent tests of GR

CHALLENGE

- QNMs are not orthogonal (they have similar oscillatory behaviors)



$$\langle v_{j,\ell mn} | v_{j',\ell' m' n'} \rangle \neq \delta_{jj'} \delta_{\ell\ell'} \delta_{mm'} \delta_{nn'}$$

Noise-weighted inner product

$$\langle A | B \rangle = \sum_{ij} A(t_i) C_{ij}^{-1} B(t_j)$$

- C_{ij} : Noise covariance matrix

Correlations between QNM amplitudes arise and **obscure the identification of individual modes.**

[E. Coleman and E. Finch \(2025\)](#)

An analysis method based on **orthonormal QNMs**

[S. Morisaki, H. Motohashi, **MS**, and D. Watarai, PRD **112**, 124083 \(2025\)](#)

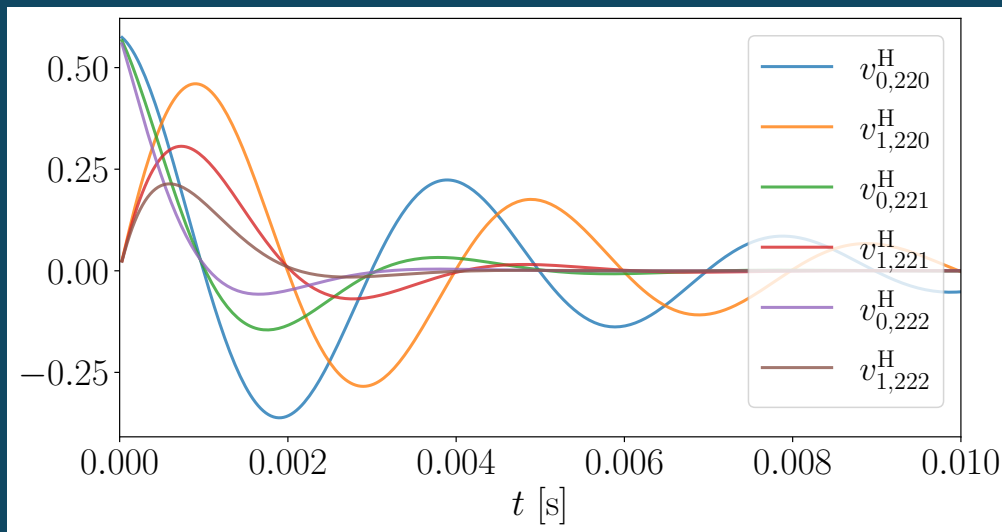
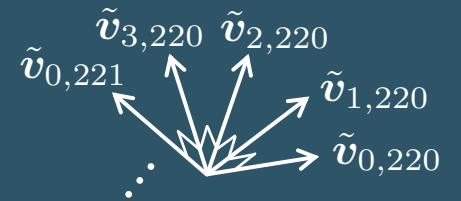
Original template waveform

$$h^I(t) = \sum_{\ell mn} \sum_{j=0}^3 c_{j,\ell mn} v_{j,\ell mn}^I(t)$$

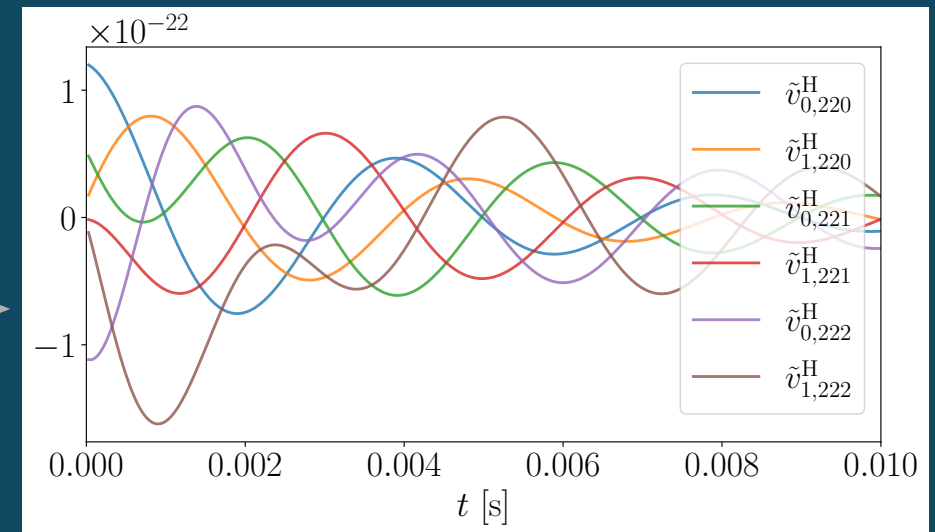


Orthonormalized template waveform

$$h^I(t) = \sum_{\ell mn} \sum_{j=0}^3 \tilde{c}_{j,\ell mn} \tilde{v}_{j,\ell mn}^I(t)$$



Numerically
orthonormalize
(Gram-Schmidt)



$$\langle v_{j,\ell mn} | v_{j',\ell' m' n'} \rangle \neq \delta_{jj'} \delta_{\ell\ell'} \delta_{mm'} \delta_{nn'}$$

$$\langle \tilde{v}_{j,\ell mn} | \tilde{v}_{j',\ell' m' n'} \rangle = \delta_{jj'} \delta_{\ell\ell'} \delta_{mm'} \delta_{nn'}$$

Correlations between QNM amplitudes are reduced.

Mode Identification

6/16

Physical-mode
amplitude

$$A_{lmn} = \sqrt{\sum_{j=0}^{D-1} c_{j,lmn}^2}$$

vs.

Orthonormal-mode
amplitude

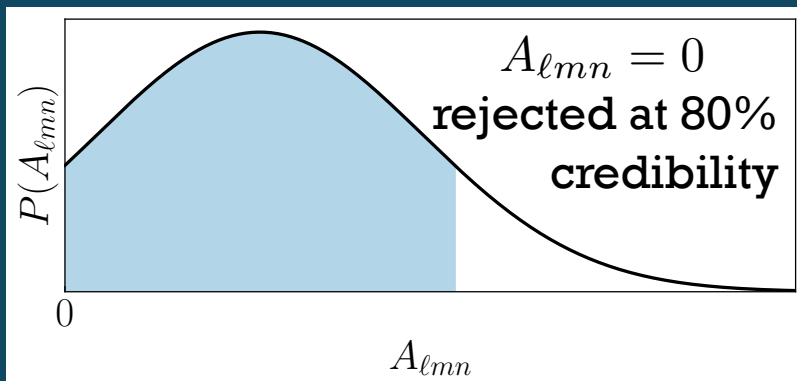
$$\tilde{A}_{lmn} = \sqrt{\sum_{j=0}^{D-1} \tilde{c}_{j,lmn}^2}$$

- Although the values of A_{lmn} and $\tilde{A}_{lmn}$ differ, they agree on whether it is zero or nonzero:

$$A_{lmn} = 0 \iff \tilde{A}_{lmn} = 0 \quad \text{and} \quad A_{lmn} \neq 0 \iff \tilde{A}_{lmn} \neq 0$$

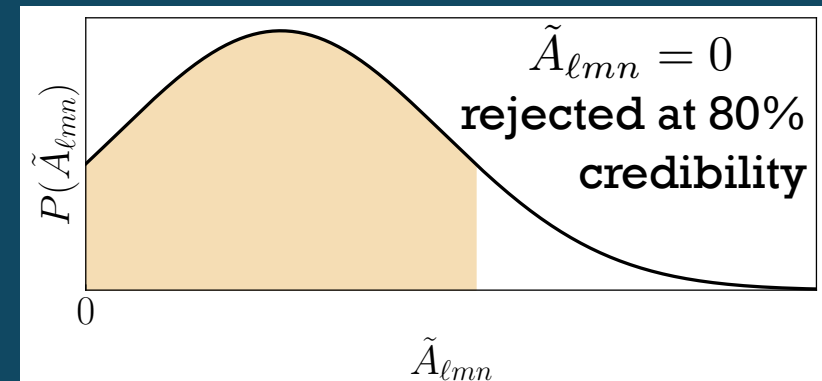
A standard approach

- The existence of a given mode is inferred from how strongly the posterior $P(A_{lmn})$ excludes $A_{lmn} = 0$ under a flat prior on A_{lmn} (or $|\mathcal{A}_{lmn}| + |\mathcal{A}'_{lmn}|$).



Analysis method based on orthonormal QNMs

- Mode existence is inferred from how strongly the 1D posterior $P(\tilde{A}_{lmn})$ disfavors $\tilde{A}_{lmn} = 0$ under a flat prior on $\tilde{A}_{lmn}$.



Application to GW250114

7/16

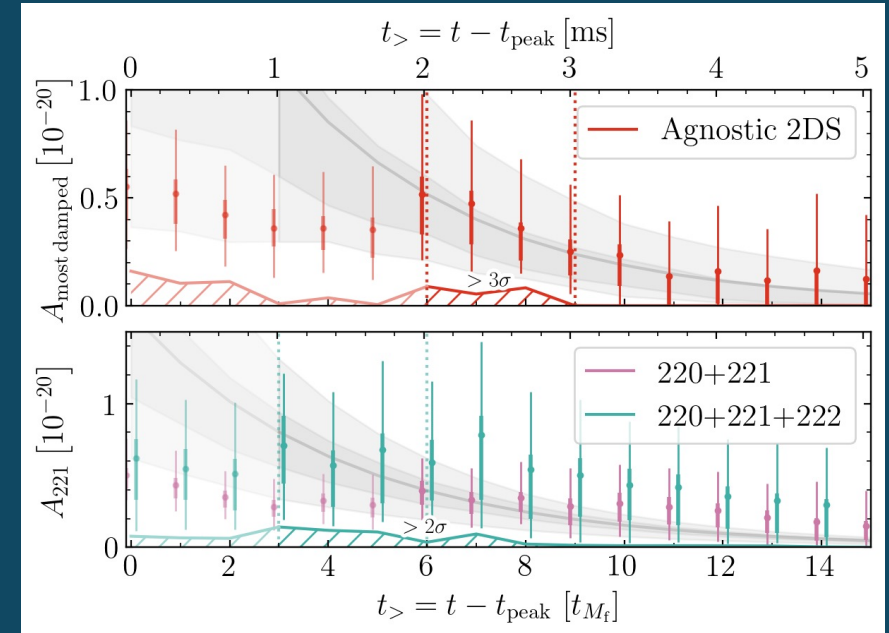
We apply our method to GW250114.

The loudest event so far (SNR~80)

[MS et al., arXiv:2605.03576 \(2026\)](#)

□ GW250114 companion paper

➤ The **(2,2,1) mode amplitude is bounded away from zero** at $6t_{M_f}$ after the peak and is **consistent with exponential decay**.



[LVK Collab. \(2025\) \(companion paper\)](#)

□ Following the previous studies, we perform the analysis with

➤ 220+221 model

➤ 220+221+222 model

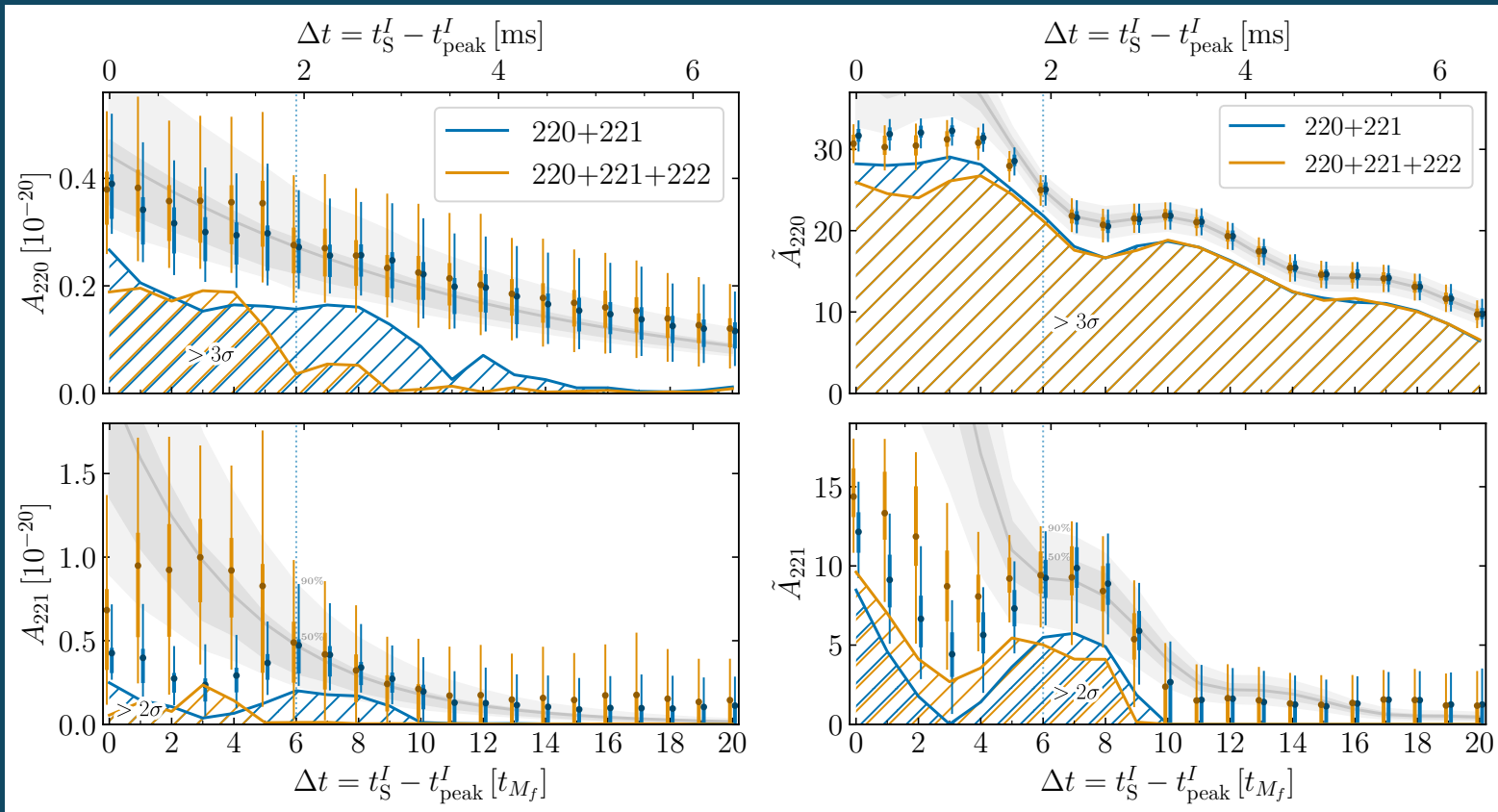
□ Following the GW250114 detection and companion papers, we adopt the same reference parameters and data conditioning.

Nonorthogonal QNMs

using the RINGDOWN package

[M. Isi & W. Farr \(2021\)](#)

Orthonormal QNMs



Nonorthogonal QNMs

- A_{221} shows a decaying behavior for $\Delta t \geq 6 t_{M_f}$.
- At early times in this regime, the **220+221 model** excludes $A_{221} = 0$ at 2σ , whereas the **220+221+222 model** does not.

Orthonormal QNMs

- Even the **220+221+222 model** excludes $\tilde{A}_{221} = 0$ at 2σ in this regime.

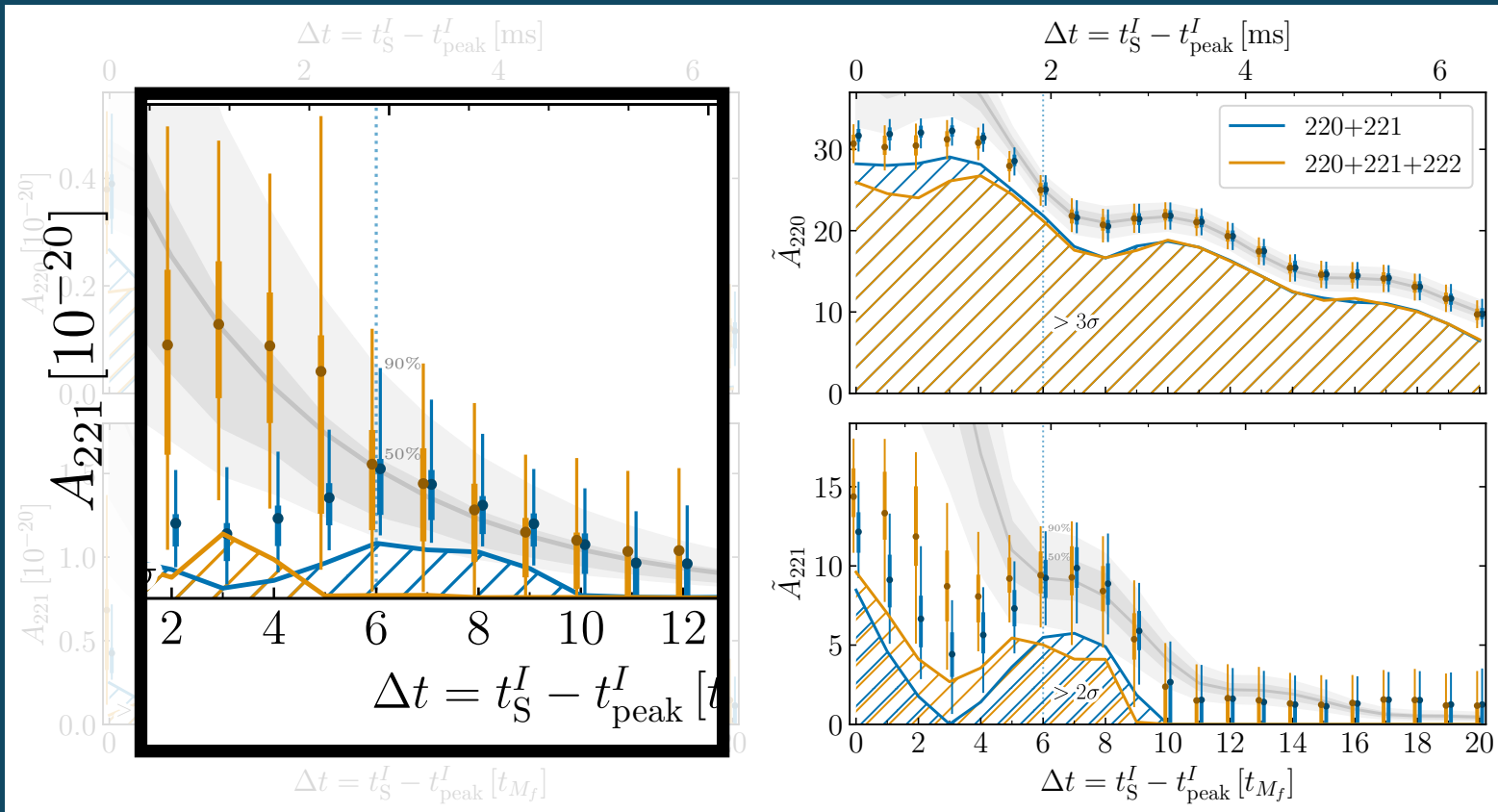
This difference is due to the **reduced correlation** between the 221 and 222 modes after orthonormalization.

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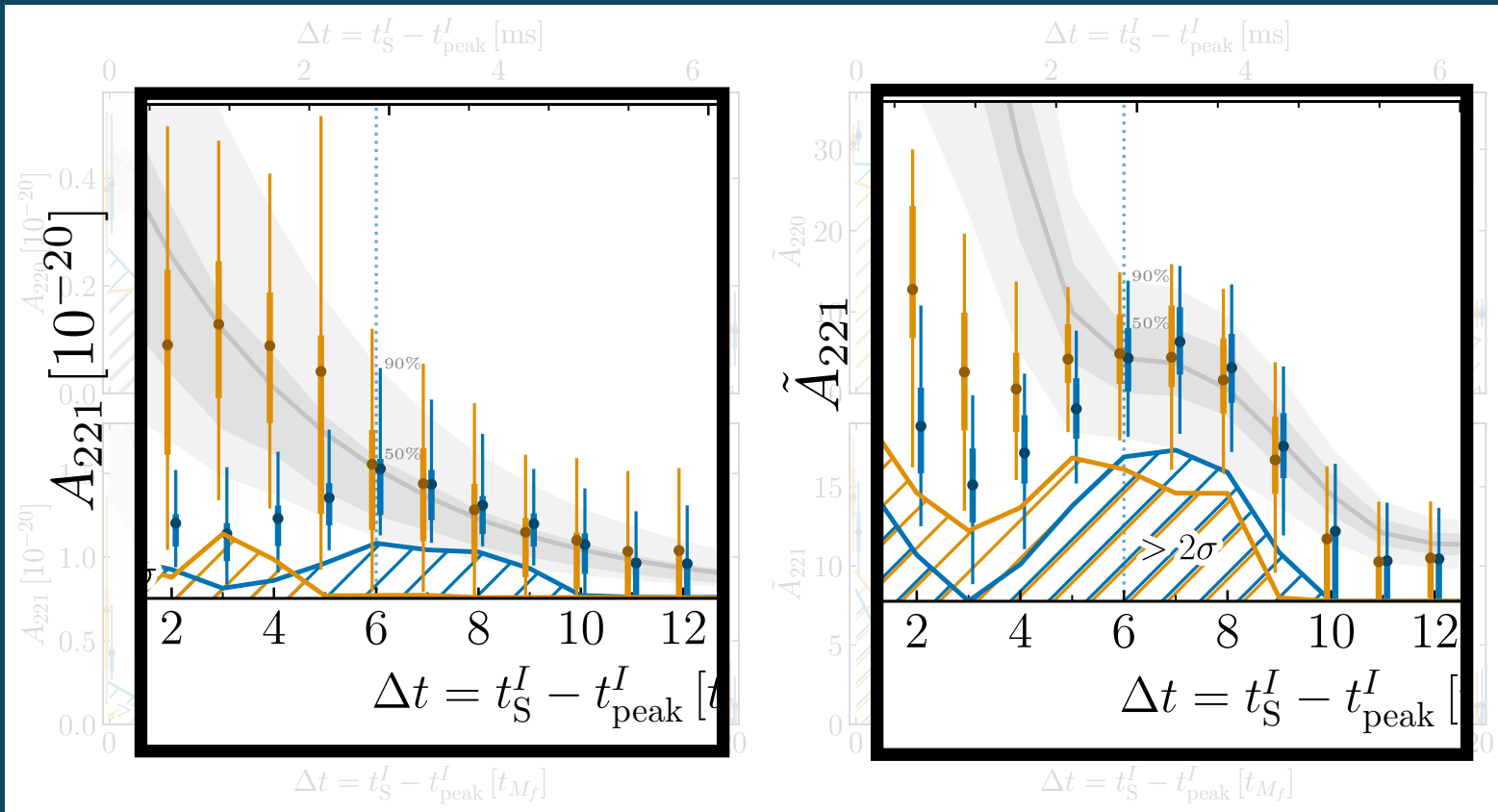
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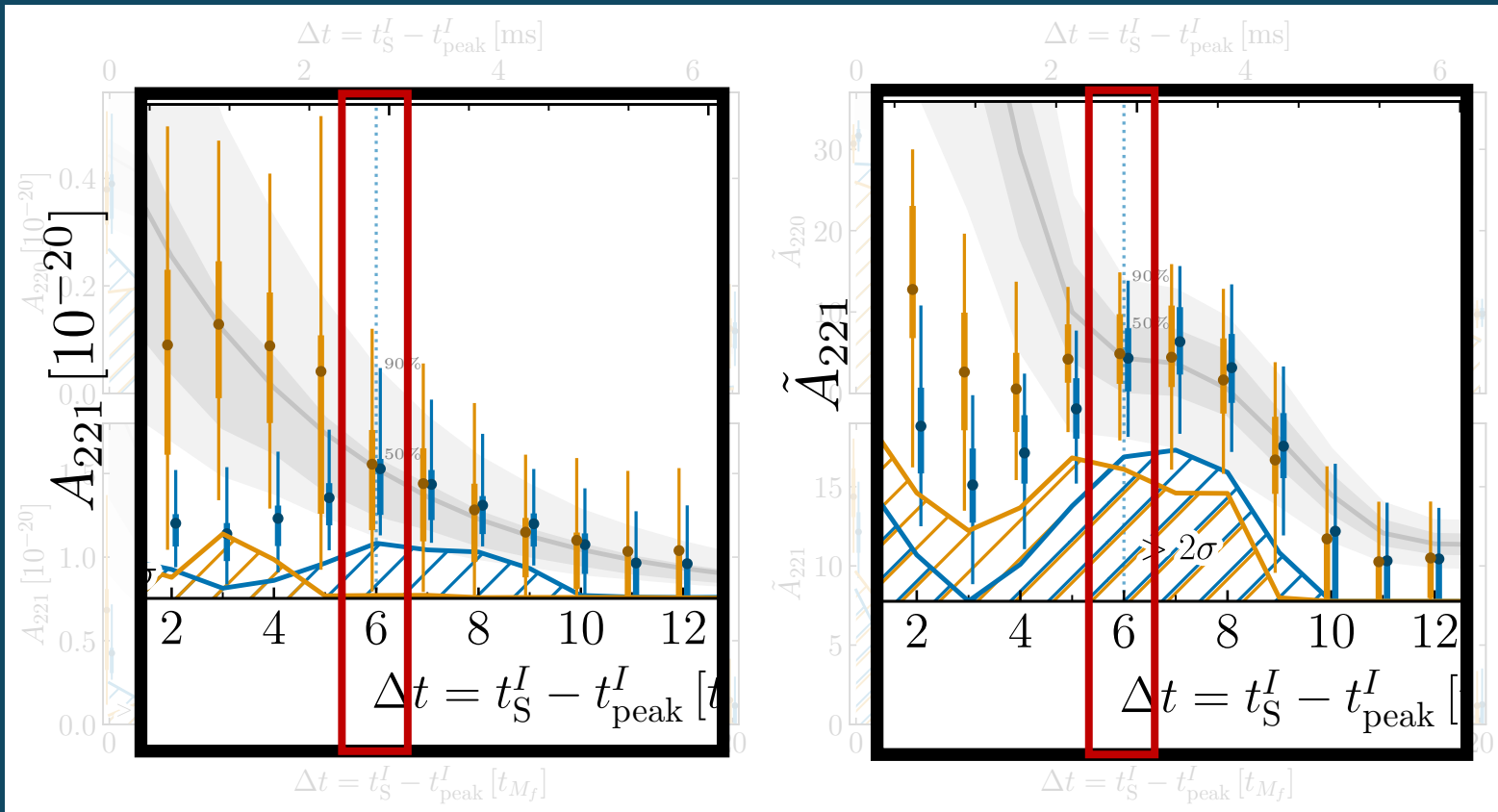
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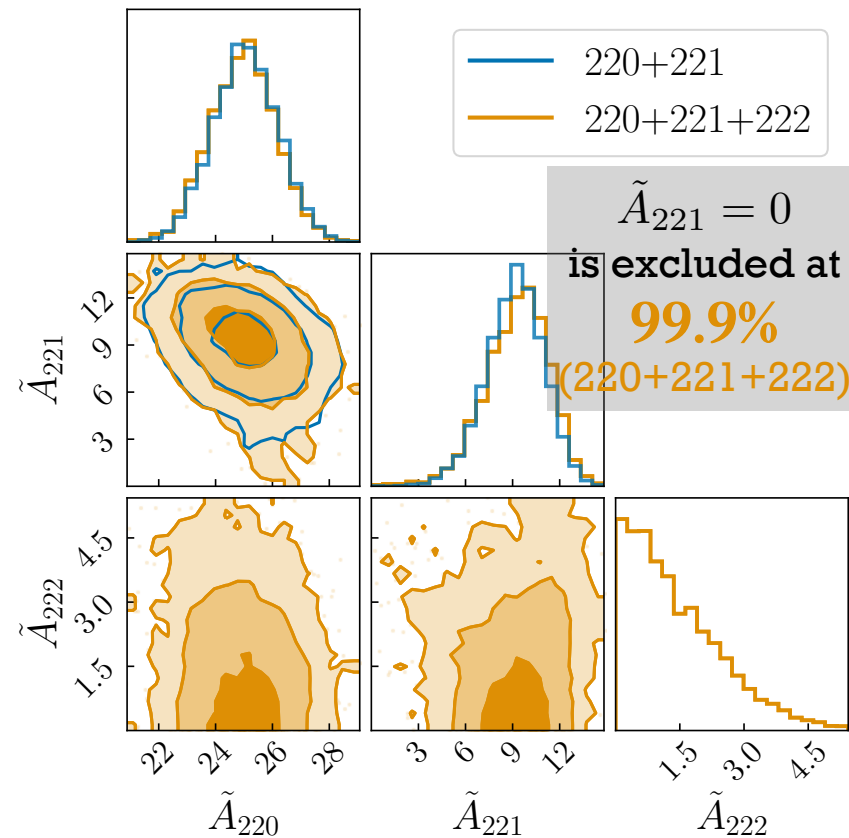
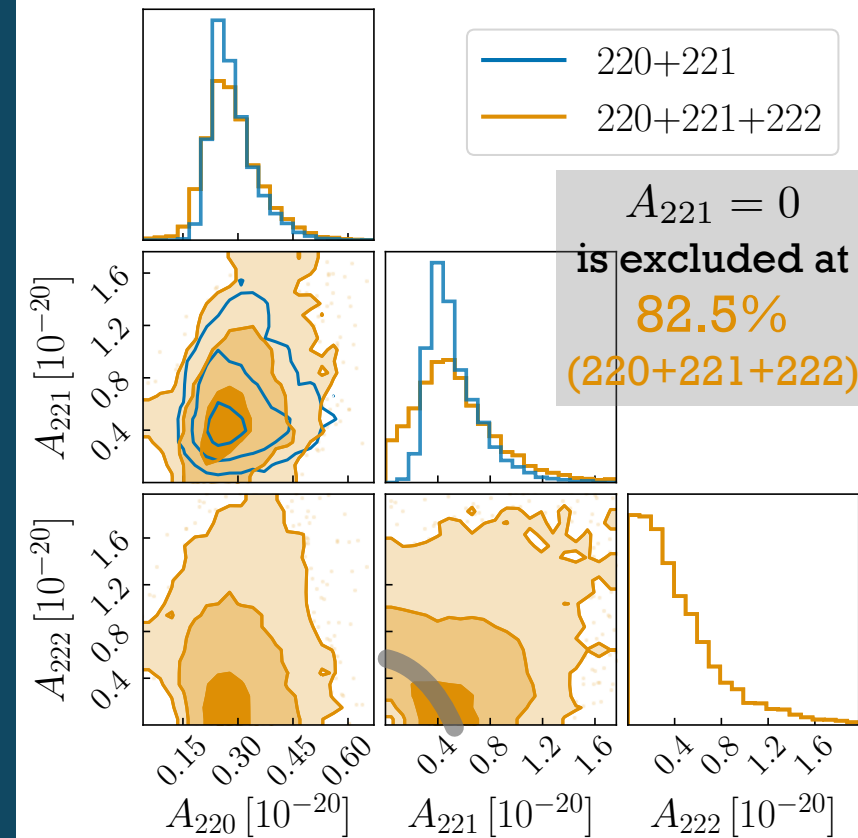
Application to GW250114

12/16

The analysis starting at $\Delta t = 6 t_{M_f}$

Nonorthogonal QNMs

Orthonormal QNMs



Nonorthogonal QNMs

- The **220+221+222 model** does not clearly exclude $A_{221} = 0$ due to the correlation between the 221 and 222 modes.

Orthonormal QNMs

- After orthonormalization, the **220+221+222 model** excludes $\tilde{A}_{221} = 0$ at the **99.9% confidence**.

The orthonormal QNM method provides a **more robust detection of QNMs**, regardless of the number of modes included in the template waveform.

Application to an Avoided-Crossing Waveform 13/16

What is avoided crossing (resonant excitation)?

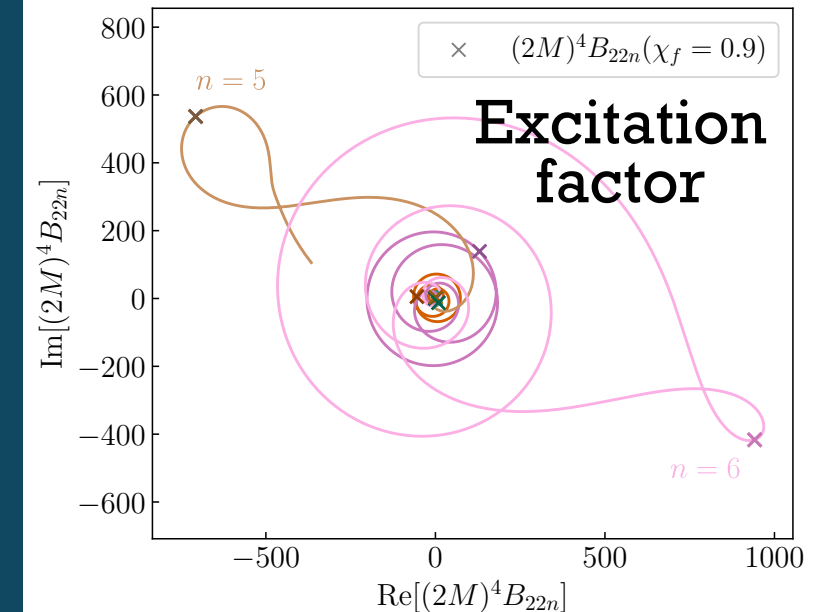
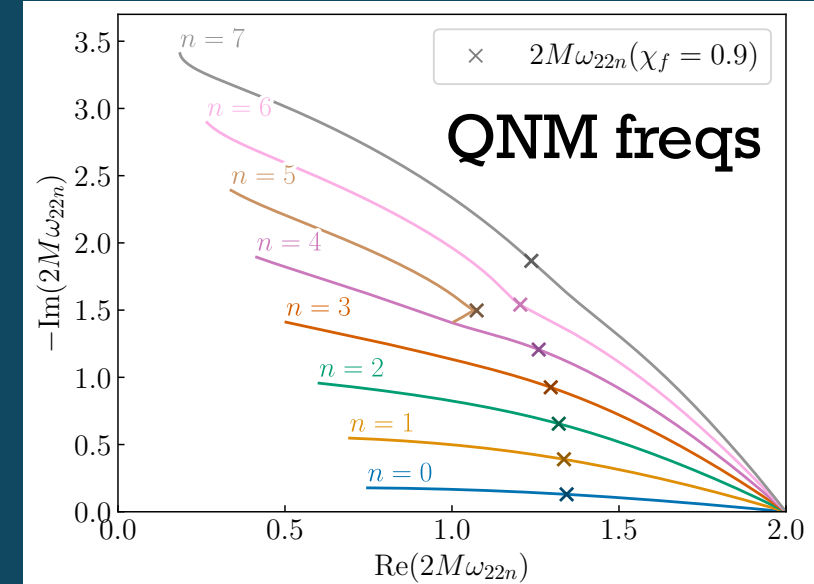
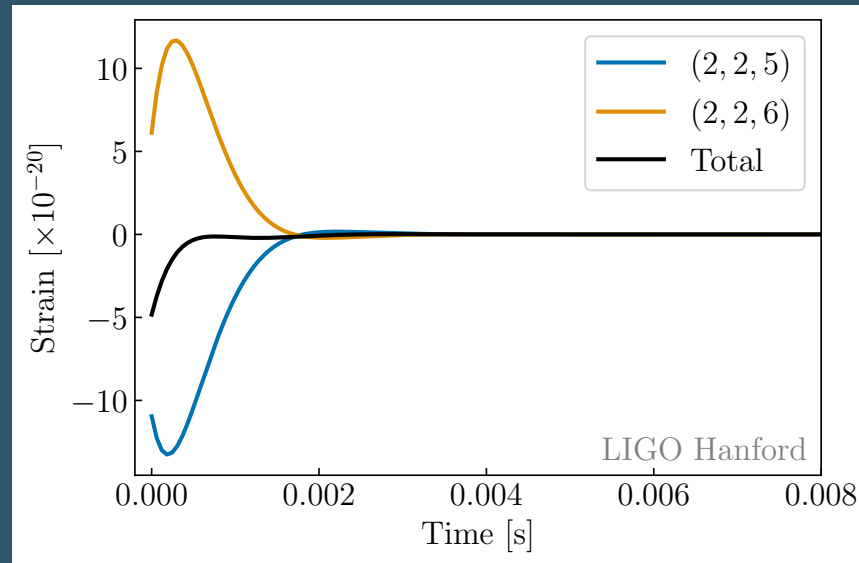
- Recently, it has been theoretically predicted that the frequencies of certain overtones (e.g., 225 and 226) become **closely spaced at high spin**, and these overtones are **resonantly excited**.

- Injection:

$$h(t) = \sum_{n=5,6} \left[\frac{B_{22n}}{(\omega_{22n})^2} e^{-i\omega_{22n}t} + \left(\frac{B_{22n}}{(\omega_{22n})^2} \right)^* e^{i\omega_{22n}^*t} \right]$$

with $M_f = 70 M_\odot$, $\chi_f = 0.9$, SNR=300.

- Analysis: 2DSs, **agnostic** to the Kerr spectrum



Application to an Avoided-Crossing Waveform 14/16

Agnostic search with the constraint $f_0 > f_1$

Nonorthogonal QNMs

Orthonormal QNMs

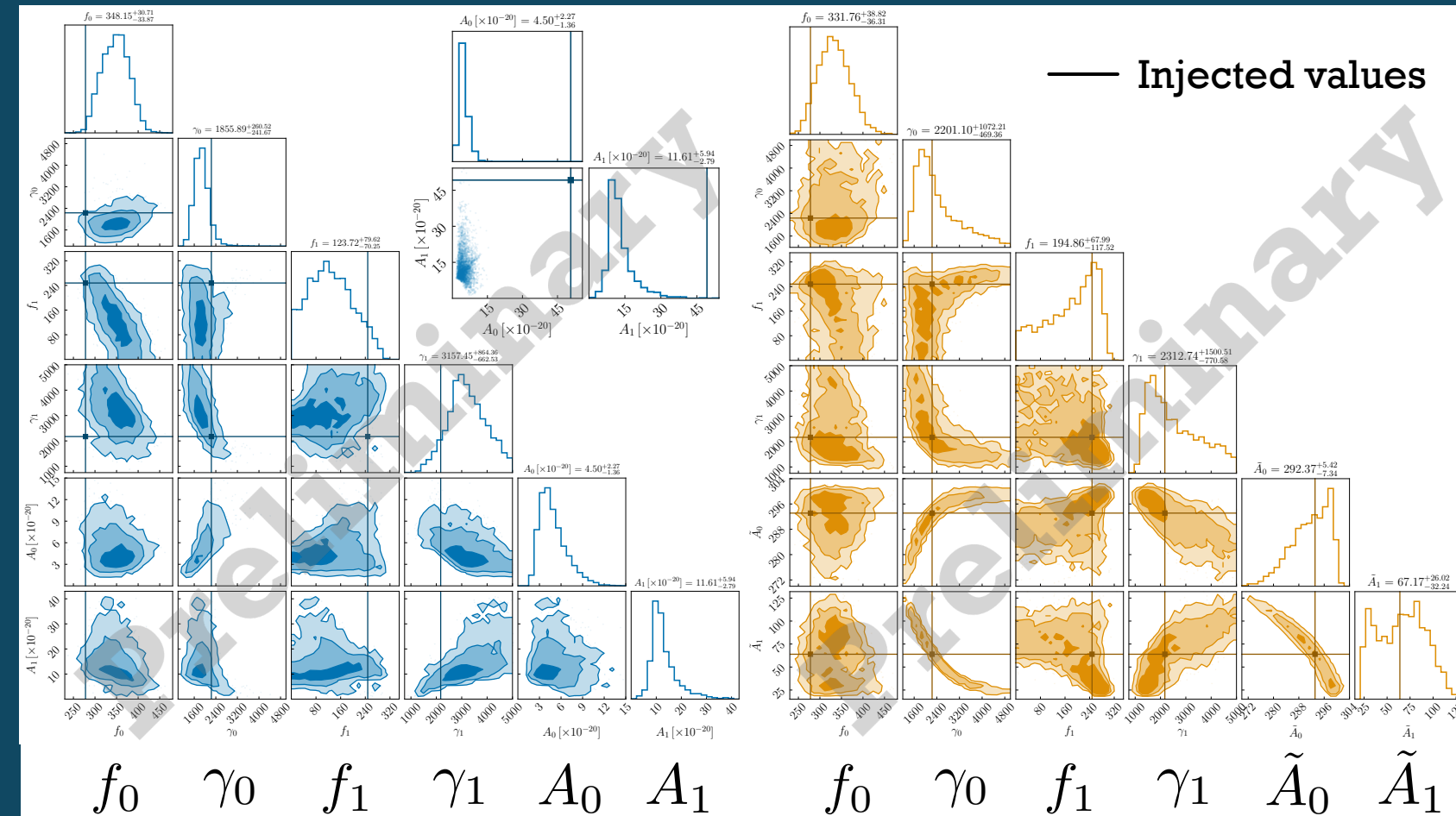
0: Highest-frequency mode
1: Lowest-frequency mode

Nonorthogonal QNMs

- The posterior peaks are **shifted away from the injected values**.
- In particular, the inferred **mode amplitudes are significantly smaller than the injected values**.

Orthonormal QNMs

- Injected values are recovered within 1σ – 2σ .



Reconstructed waveform at Hanford

Nonorthogonal QNMs

Orthonormal QNMs

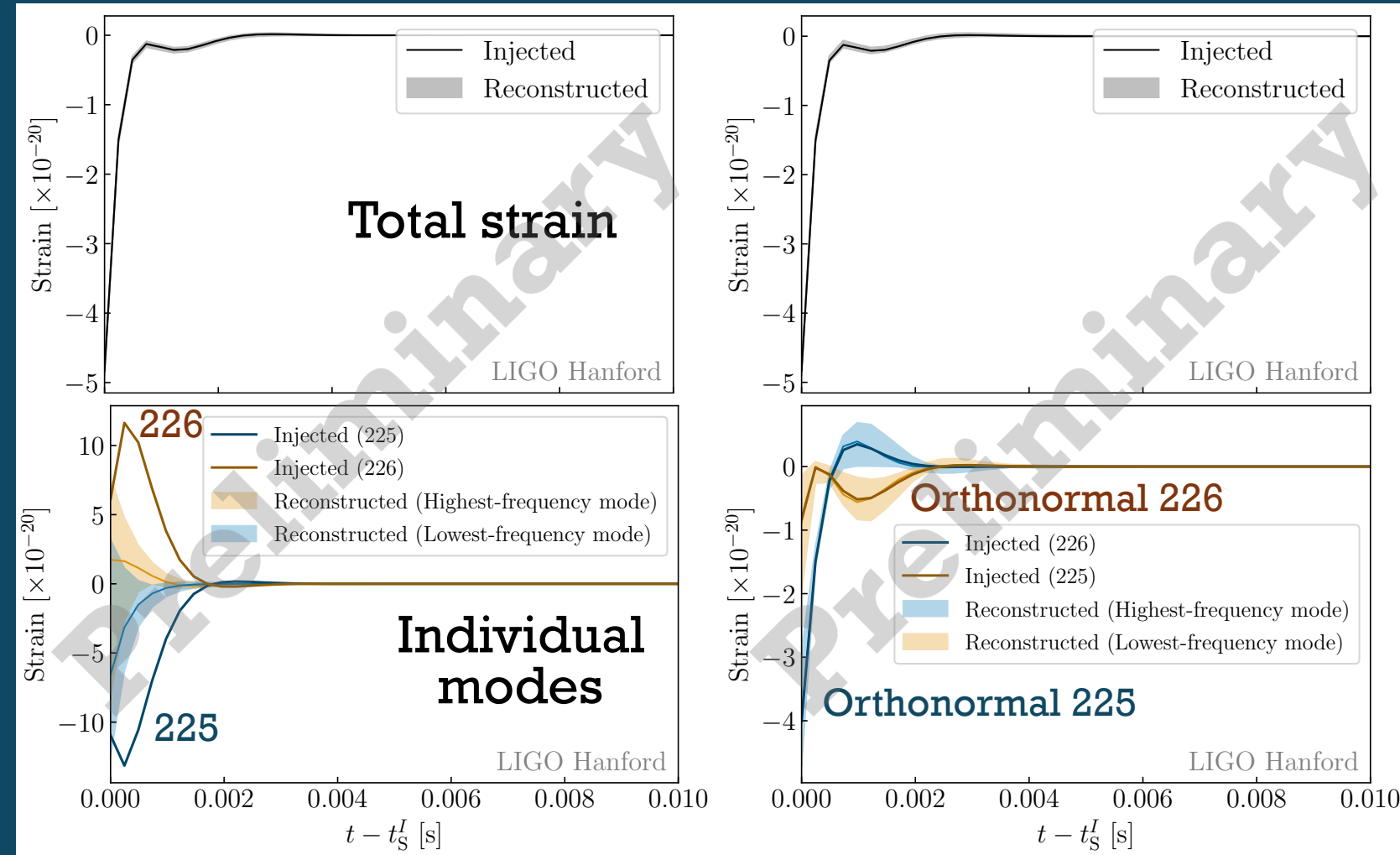
Nonorthogonal QNMs

- Total strain recover the injected waveform.
- Individual modes are underestimated and weakly constrained.

Orthonormal QNMs

- Both the total strain and individual modes recover the injected waveforms.

The orthonormal QNM method is highly effective for **extracting individual modes** from strongly cancelling waveforms.



- The orthonormal QNM method **reduces correlations** between QNM amplitudes.
- It finds **stronger evidence for the (2,2,1) mode** in the 220+221+222 model compared to the conventional method in GW250114.
- It provide a powerful way to extract individual modes from strongly cancelling waveforms (e.g., avoided crossings)
- This method will be **more powerful in future observations** (O5, ET, CE), as more QNMs are expected to be detected.